Session 6: Cable Damage and Repair

Case Study Breakout Session

nodal/ys

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The Resilience Cycle

Prevent

Protection,

Risk reduction

Detect

Monitoring, Threat alerts,

Notification

Respond

Address threat, Permitting,

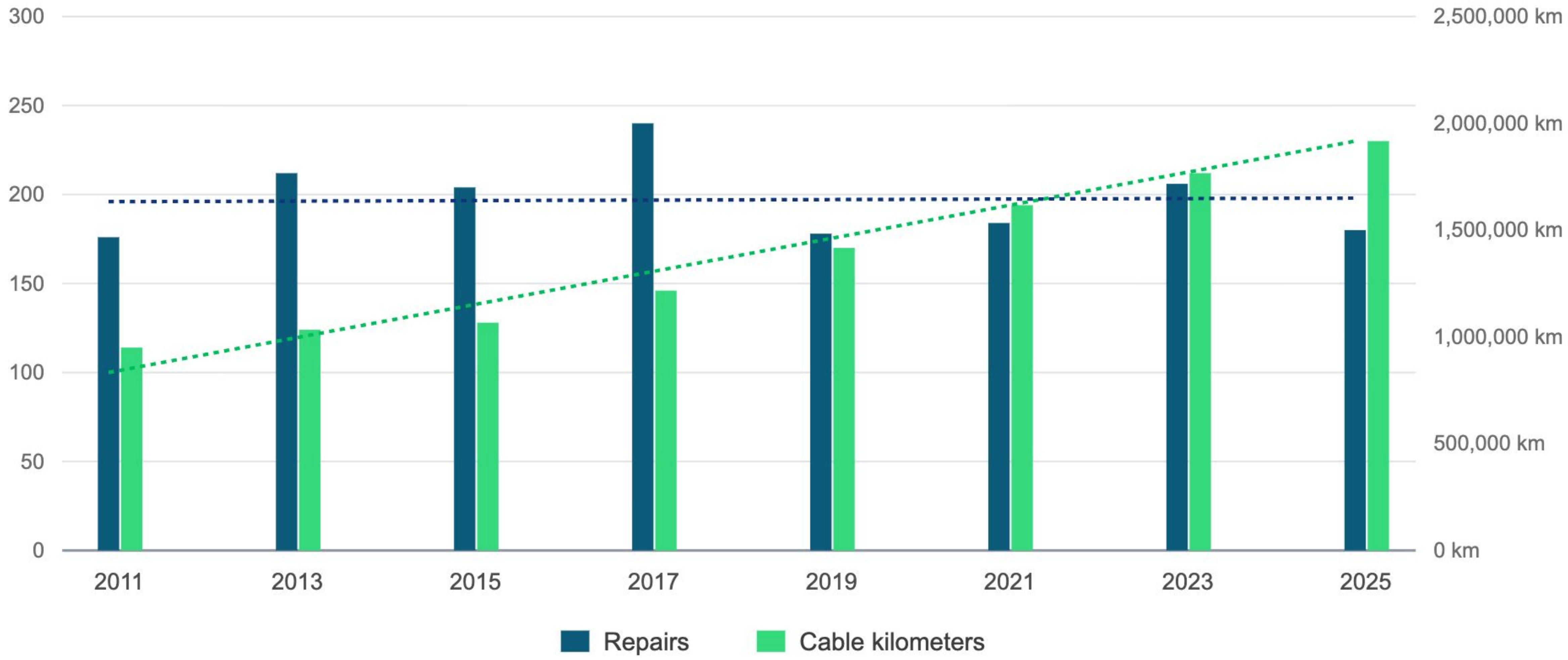
Mobilise repair vessel

Recover

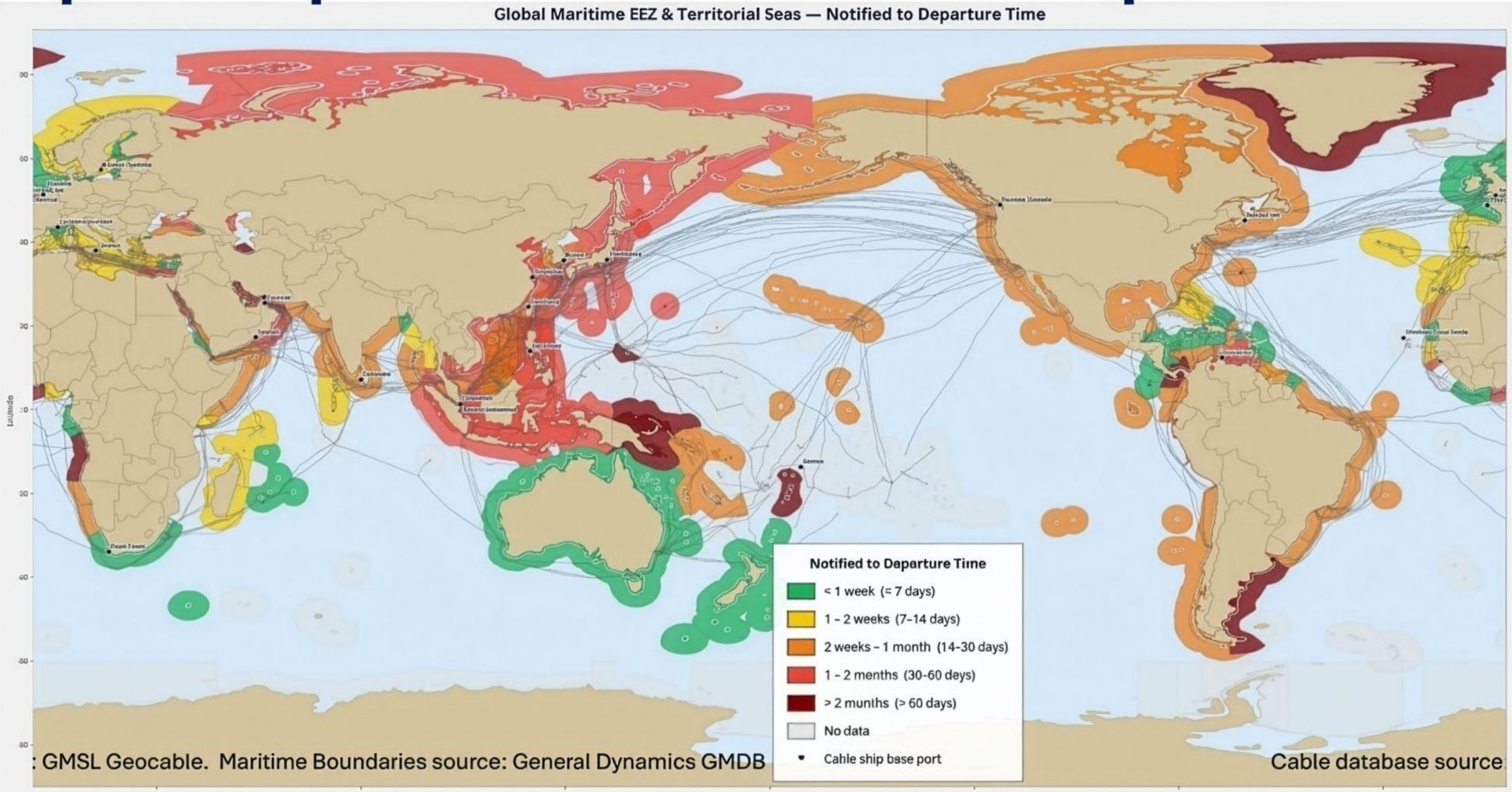
Restore service,

Share and integrate learnings

Repairs per year vs. total in-service cable length

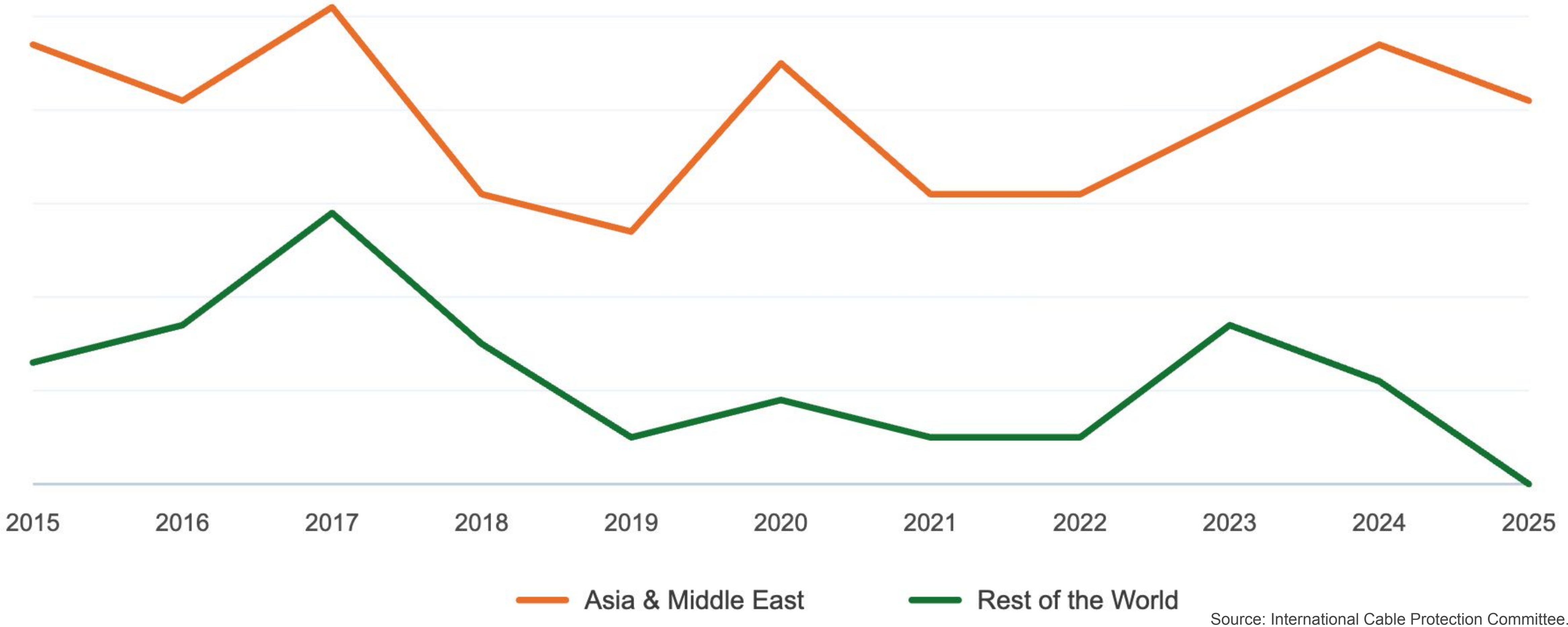


Repair Response Time: Notified to Departure



Source: International Cable Protection Committee. Modified to Asia-Pacific map centre by Nodalys.

Asia-Middle East repair frequency vs. rest of the world



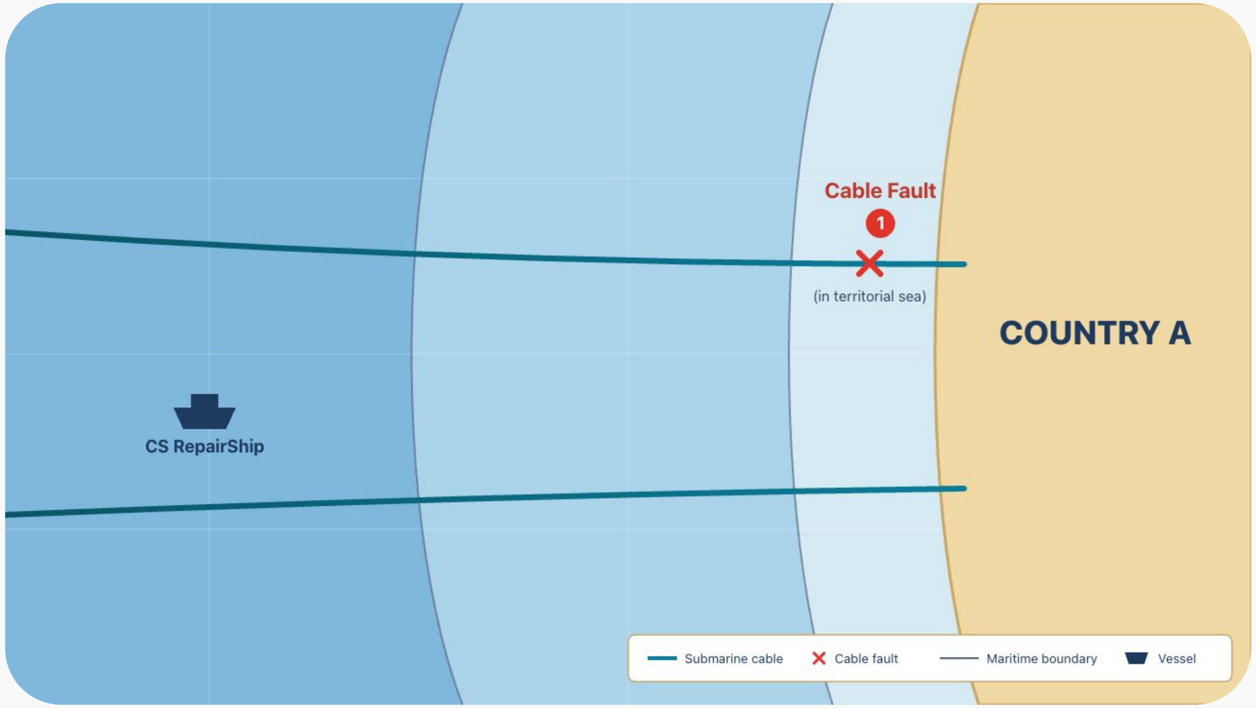
Source: International Cable Protection Committee.

Scenario 1: Cable Fault (in territorial sea)

Country A's submarine cable has been cut within its territorial sea by a ship anchor. It's causing internet slowdowns for businesses and millions of everyday users.

The **CS RepairShip** can perform the repair. It is flagged to Country Z.

The vessel can be ready to sail within 24 hours, but **the repair ship operator has some regulatory processes to go through first and will not mobilise until they know they are able to get to the repair site**, fearful of legal penalties or military interception.



Scenario 1 continued

The repair operator encounters the following:

a) Agency ping-pong — 3 days

The operator tries to work out which government ministry is responsible. They get passed around through 4 ministries.

b) Foreign ship in domestic waters — 7 days

Cable repair falls under cabotage laws. The ship is foreign-flagged. Country A's laws restrict foreign ships and crew from working in domestic waters.

c) Equipment import duties — 7 days

Country A treats spare cables and repeaters as standard commercial imports. Equipment held at port while duties are assessed. Disagreements on item values.

d) Long processing times — 43 days

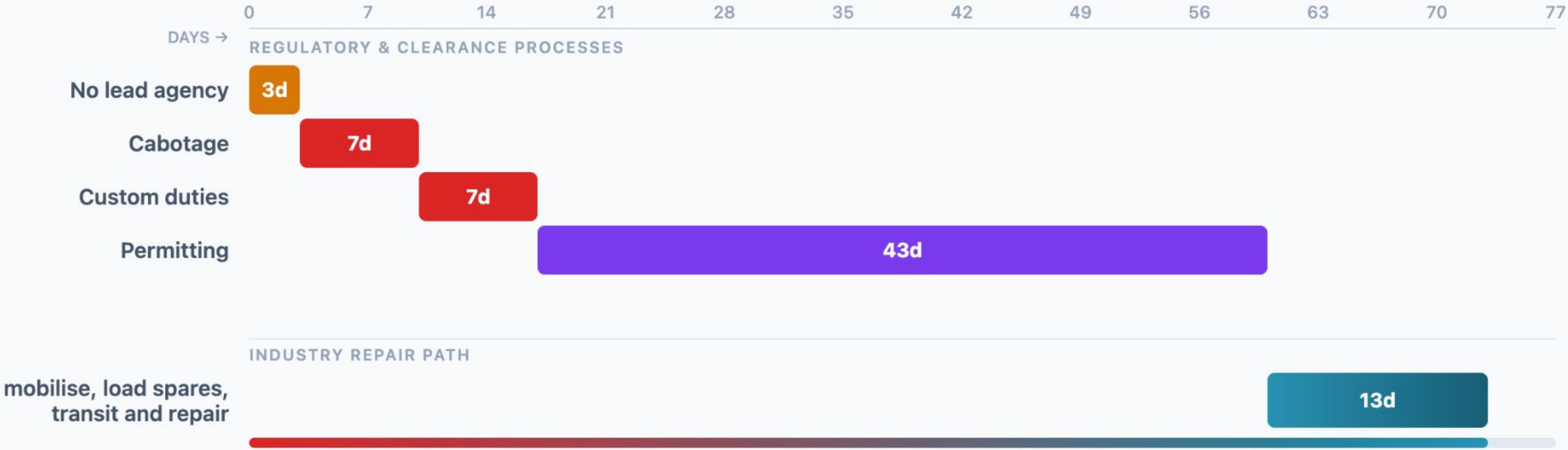
Applications for environmental, spatial, maritime, and security permits. Some are sequential, managed by different departments.

The Repair Timeline

INDUSTRY
13
days

GOVERNMENT
+60
days

TOTAL TO REPAIR
73
days



You are members of the Country A Cable Repair Coordination Forum

The Forum convenes government **regulators, cable owners, and operators to facilitate cross-stakeholder coordination and harmonisation of regulations and procedures for cable repairs**, including technical and logistical aspects, with the aim of addressing issues and providing solutions to support submarine cable maintenance and repair.

Question 1

How would you make the processes (a) - (d) faster and more streamlined so the repair ship can start fixing the fault sooner?

a) Agency ping-pong (3 days): The operator tries to work out which government ministry is responsible. They get passed around through 4 ministries.

b) Foreign ship in domestic waters (7 days): Cable repair falls under cabotage laws. The ship is foreign-flagged. Country A's laws restrict foreign ships and crew from working in domestic waters.

c) Equipment import duties (7 days): Country A treats spare cables and repeaters as standard commercial imports. Equipment held at port while duties are assessed. Disagreements on item values.

d) Long processing times (43 days): Applications for environmental, spatial, maritime, and security permits. Some are sequential, managed by different departments.

⚠ SECOND FAULT — Network outage
Estimated impact: 200M per day

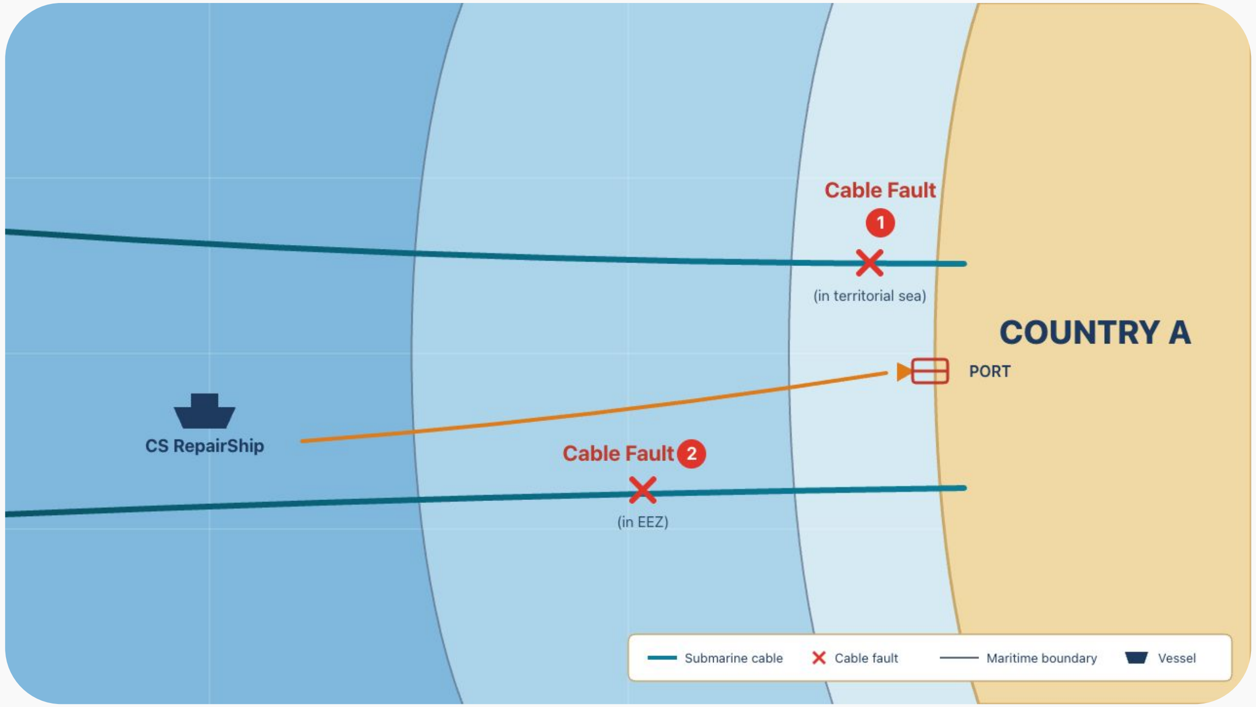
Scenario 1.5: Cable Fault (in EEZ)

A fishing trawler has just cut through Country A's second submarine cable in Country A's EEZ. Both cables are now down. The internet is practically unusable.

e) Port entry for cable fault in EEZ (3 days):

Despite the fault being in the EEZ, Country A requires the vessel to make a detour and dock at port to clear immigration, pay port fees, and formally process the temporary importation of the subsea cable equipment spares.

This adds 3 days in addition to the other processes in Question 1.



Question 1 & 2

How would you make the processes (a) - (e) faster and more streamlined so the repair ship can start fixing the fault sooner?

a) Agency ping-pong (3 days): The operator tries to work out which government ministry is responsible. They get passed around through 4 ministries.

b) Foreign ship in domestic waters (7 days): Cable repair falls under cabotage laws. The ship is foreign-flagged. Country A's laws restrict foreign ships and crew from working in domestic waters.

c) Equipment import duties (7 days): Country A treats spare cables and repeaters as standard commercial imports. Equipment held at port while duties are assessed. Disagreements on item values.

d) Long processing times (43 days): Applications for environmental, spatial, maritime, and security permits. Some are sequential, managed by different departments.

e) Port entry for cable fault in EEZ (3 days): Despite the fault being in the EEZ, Country A requires the vessel to make a detour and dock at port to clear immigration, pay port fees, and formally process the temporary importation of the subsea cable equipment spares.

Question 3: Reflecting on your own country's experience...

Has your country adopted or tried to adopt any practices to streamline or expedite cable repairs? If so...

a) Motivation

What were the key triggers that motivated change?

b) Challenges

What challenges were faced, and how were they overcome?

c) Ongoing

What challenges do you continue to face?